50 MCQs: Layer 2 Switching

Easy (10)

1.

   At which OSI layer do switches operate?
   A) Layer 1
   B) Layer 2
   C) Layer 3
   D) Layer 4
   Answer: B
   Explanation: Switches operate at the Data Link layer (Layer 2), forwarding frames based on MAC addresses.

2.

   What does a Layer 2 switch use to forward frames?
   A) IP addresses
   B) MAC addresses
   C) Port numbers
   D) Subnet masks
   Answer: B
   Explanation: Layer 2 switches forward frames based on destination MAC addresses.

3.

   What does the MAC address table in a switch store?
   A) IP address to port mapping
   B) MAC address to port mapping

C) VLAN IDs
D) Routing tables
Answer: B
Explanation: Switches map MAC addresses to specific ports to forward frames correctly.

4.

What happens when a switch receives a frame with an unknown destination MAC address?
A) Drops the frame
B) Broadcasts to all ports except incoming
C) Sends an ICMP message
D) Sends the frame back to sender
Answer: B
Explanation: The switch floods unknown destination frames to all ports except the one it came from.

5.

Which device creates multiple collision domains?
A) Hub
B) Switch
C) Router
D) Modem
Answer: B
Explanation: Switches separate each port into its own collision domain.

6.

What is the purpose of Spanning Tree Protocol (STP)?
A) Speed up routing
B) Prevent Layer 2 loops
C) Manage IP addresses
D) Provide encryption
Answer: B

Explanation: STP prevents loops that cause broadcast storms in Layer 2 networks.

7.

What is the broadcast MAC address?
A) 00:00:00:00:00:00
B) FF:FF:FF:FF:FF:FF
C) 01:00:5E:00:00:00
D) 11:22:33:44:55:66
Answer: B
Explanation: FF:FF:FF:FF:FF:FF is the broadcast MAC address used to send frames to all devices in a LAN.

8.

Which switching method checks the entire frame for errors before forwarding?
A) Cut-Through
B) Store-and-Forward
C) Fragment-Free
D) Fast-Forward
Answer: B
Explanation: Store-and-Forward switches receive the whole frame, check CRC, and forward only if error-free.

9.

VLANs are used to:
A) Increase collision domains
B) Increase broadcast domains
C) Separate broadcast domains
D) Combine IP networks
Answer: C
Explanation: VLANs logically separate a switch into multiple broadcast domains.

10.

A trunk link on a switch carries:
A) Traffic for a single VLAN
B) Traffic for multiple VLANs
C) Only broadcast traffic
D) Only management traffic
Answer: B
Explanation: Trunk links carry traffic from multiple VLANs, using tagging protocols like 802.1Q.

Medium (30)

11.

Which protocol is used for VLAN tagging in most modern switches?
A) ISL
B) 802.1Q
C) MPLS
D) STP
Answer: B
Explanation: IEEE 802.1Q is the standard protocol for VLAN tagging in Ethernet frames.

12.

What is the default aging time for MAC address entries in a Cisco switch?
A) 60 seconds
B) 180 seconds
C) 300 seconds
D) 600 seconds

Answer: C
Explanation: MAC addresses age out after 300 seconds (5 minutes) if no frames are received.

13.

Which switching method minimizes latency by forwarding frames immediately after reading the destination MAC?
A) Store-and-Forward
B) Cut-Through
C) Fragment-Free
D) Fast-Forward
Answer: B
Explanation: Cut-Through forwarding starts transmitting once the destination MAC is read, lowering latency but risking forwarding corrupt frames.

14.

Which command shows the MAC address table on a Cisco switch?
A) show ip route
B) show mac address-table
C) show vlan
D) show interfaces
Answer: B
Explanation: show mac address-table displays the MAC address to port mappings.

15.

What is the function of a Designated Port in STP?
A) Blocks traffic
B) Forwards frames on a LAN segment
C) Disables the port
D) Manages VLAN assignments
Answer: B
Explanation: The Designated Port forwards frames on a LAN segment in STP topology.

16.

What happens during a broadcast storm?
A) Frames are delayed
B) Broadcast traffic overwhelms the network
C) Routing tables get corrupted
D) Switch drops all unicast frames
Answer: B
Explanation: Broadcast storms flood the network with excessive broadcast frames, causing congestion.

17.

Which of the following is NOT a characteristic of VLANs?
A) Separate broadcast domains
B) Operate at Layer 2
C) Require physical separation of devices
D) Improve security and segmentation
Answer: C
Explanation: VLANs separate networks logically without physical separation.

18.

What does enabling port security on a switch port do?
A) Limits the number of MAC addresses on a port
B) Blocks all traffic except management
C) Encrypts data on the port
D) Allows only IP-based filtering
Answer: A
Explanation: Port security limits the number and type of MAC addresses allowed on a port.

19.

Which of the following MAC addresses is a multicast address?

A) 00:00:00:00:00:01
B) FF:FF:FF:FF:FF:FF
C) 01:00:5E:00:00:01
D) 11:22:33:44:55:66
Answer: C
Explanation: MAC addresses starting with 01:00:5E are IPv4 multicast addresses.

20.

Which Layer 2 switching method forwards frames after reading the first 64 bytes?
A) Store-and-Forward
B) Cut-Through
C) Fragment-Free
D) Fast-Forward
Answer: C
Explanation: Fragment-Free waits for the first 64 bytes to avoid forwarding collision fragments.

21.

How does a switch learn MAC addresses?
A) From routing protocols
B) From the source MAC address of incoming frames
C) Manually configured only
D) From the destination IP address
Answer: B
Explanation: Switches learn MAC addresses by inspecting the source MAC of incoming frames.

22.

Which STP state does a port spend most time in during normal operation?
A) Blocking
B) Listening
C) Learning

D) Forwarding
Answer: D
Explanation: In stable networks, ports spend most time forwarding frames.

23.

What is the primary purpose of VLAN Trunking Protocol (VTP)?
A) To secure VLANs
B) To synchronize VLAN information between switches
C) To route between VLANs
D) To manage MAC address tables
Answer: B
Explanation: VTP helps propagate VLAN configurations across multiple switches.

24.

What is the effect of a MAC address table overflow attack?
A) Switch forwards frames only to correct ports
B) Switch floods all frames out every port, behaving like a hub
C) Switch drops all incoming frames
D) Switch resets
Answer: B
Explanation: Overflowing MAC tables causes the switch to flood frames, enabling sniffing.

25.

Which command disables STP on a specific switch port?
A) no spanning-tree
B) spanning-tree portfast
C) shutdown
D) spanning-tree disable
Answer: B (PortFast enables immediate forwarding but doesn't fully disable STP)
Explanation: spanning-tree portfast speeds up port transition but STP still runs; full disable is not recommended.

26.

In 802.1Q VLAN tagging, how many bytes are added to the Ethernet frame?
A) 2
B) 4
C) 8
D) 12
Answer: B
Explanation: 802.1Q inserts a 4-byte VLAN tag into the Ethernet frame header.

27.

Which VLAN ID is reserved for the native VLAN on 802.1Q trunks by default?
A) 0
B) 1
C) 100
D) 4095
Answer: B
Explanation: VLAN 1 is the default native VLAN on Cisco switches.

28.

What is a collision domain?
A) Area where data packets collide due to shared medium
B) Logical segmentation of a network
C) Area where broadcasts are contained
D) Group of VLANs
Answer: A
Explanation: A collision domain is a network segment where devices compete for the same bandwidth.

29.

How does a switch handle a frame with a multicast MAC address?

A) Drops the frame
B) Forwards to all ports except incoming
C) Forwards to specific ports with multicast group members
D) Converts it to a broadcast frame
Answer: C
Explanation: Multicast frames are forwarded only to ports that belong to the multicast group.

30.

Which of the following is true about store-and-forward switching?
A) Forwards frames immediately without error checking
B) Waits for entire frame and checks CRC before forwarding
C) Forwards only fragments of frames
D) Uses the first 14 bytes to decide forwarding
Answer: B
Explanation: Store-and-forward checks the entire frame for errors before forwarding.

31.

Which switching method can forward corrupted frames?
A) Store-and-Forward
B) Cut-Through
C) Fragment-Free
D) All of the above
Answer: B
Explanation: Cut-Through switches forward frames before CRC checking, so corrupted frames may be forwarded.

32.

Which VLAN configuration mode allows a port to carry traffic from multiple VLANs?
A) Access mode
B) Trunk mode
C) Routed mode

D) Monitor mode
Answer: B
Explanation: Trunk mode ports carry multiple VLANs using tagging.

33.

What is the maximum number of VLANs supported in 802.1Q?
A) 64
B) 128
C) 4094
D) 65536
Answer: C
Explanation: 802.1Q supports up to 4094 VLANs.

34.

What happens if two switches connected by a trunk link have different native VLANs configured?
A) No issues
B) VLAN mismatch causing traffic loss on native VLAN
C) STP disables the port
D) Trunk link drops all traffic
Answer: B
Explanation: Native VLAN mismatch causes untagged traffic to be dropped or misrouted.

35.

Which protocol can dynamically negotiate trunk links between switches?
A) STP
B) VTP
C) DTP
D) CDP
Answer: C
Explanation: Dynamic Trunking Protocol (DTP) negotiates trunk links automatically.

36.

Which port type on a switch forwards traffic but blocks STP BPDUs?
A) Access port
B) Trunk port
C) Edge port (PortFast)
D) Disabled port
Answer: C
Explanation: Edge ports forward traffic immediately and do not participate in STP BPDU processing.

37.

Which of the following best describes a MAC address?
A) 32-bit logical address
B) 48-bit physical address
C) 16-bit port number
D) 128-bit unique identifier
Answer: B
Explanation: MAC addresses are 48-bit hardware addresses unique to network interfaces.

38.

What is the primary advantage of using VLANs?
A) Increase collision domains
B) Separate broadcast domains
C) Combine routing domains
D) Increase IP subnet sizes
Answer: B
Explanation: VLANs separate broadcast domains, improving traffic management and security.

39.

What does the “show spanning-tree” command display?

A) Routing tables
B) VLAN configurations
C) STP status and port roles
D) MAC address table
Answer: C
Explanation: Displays Spanning Tree Protocol status, root bridge, and port states.

40.

Which Layer 2 address is used in ARP requests?
A) Destination MAC address of the target device
B) Broadcast MAC address
C) Switch MAC address
D) Router MAC address
Answer: B
Explanation: ARP requests use the broadcast MAC address to find a device's MAC from its IP.

Hard (10)

41.

How does Rapid Spanning Tree Protocol (RSTP) improve convergence time?
A) Blocks all ports immediately
B) Uses PortFast on all ports
C) Adds new port roles and rapid transition states
D) Eliminates root bridge election
Answer: C
Explanation: RSTP (802.1w) introduces new port roles and states for faster convergence.

42.

Which type of Layer 2 attack floods a switch with fake MAC addresses?
A) VLAN hopping
B) MAC spoofing
C) MAC flooding
D) ARP poisoning
Answer: C
Explanation: MAC flooding overwhelms the MAC table, causing the switch to flood all frames.

43.

In a switch with VLANs configured, which device performs routing between VLANs?
A) Layer 2 switch
B) Layer 3 switch or router
C) Hub
D) Firewall
Answer: B
Explanation: Routing between VLANs requires a Layer 3 device or Layer 3 switch.

44.

What effect does enabling BPDU Guard on a switch port have?
A) Allows the port to send BPDUs
B) Shuts down port on receipt of BPDU to protect against loops
C) Prevents VLAN hopping
D) Increases broadcast domain size
Answer: B
Explanation: BPDU Guard disables the port if a BPDU is received, protecting edge ports from STP loops.

45.

What is the default behavior of a switch when it receives a frame with a multicast MAC address it doesn’t recognize?

A) Drops the frame
B) Floods it to all ports except the source port
C) Forwards only to ports with multicast members
D) Sends an error message
Answer: B
Explanation: Unknown multicast frames are flooded to all ports except the source.

46.

What is a key difference between a Layer 2 switch and a bridge?
A) Bridges operate at Layer 3
B) Switches support multiple ports and higher speed
C) Bridges do not forward frames
D) Switches cannot separate collision domains
Answer: B
Explanation: Switches are multiport devices optimized for performance compared to bridges.

47.

How does VLAN pruning help optimize network performance?
A) Limits VLAN propagation only to switches that need them
B) Disables unused VLANs globally
C) Blocks all broadcasts
D) Aggregates VLANs into a single broadcast domain
Answer: A
Explanation: VLAN pruning restricts VLAN traffic to only trunk links where VLANs are needed.

48.

Which of the following is TRUE about the native VLAN on a Cisco switch trunk?
A) Native VLAN traffic is always tagged
B) Native VLAN traffic is sent untagged
C) Native VLAN cannot be changed

D) Native VLAN carries only management traffic
Answer: B
Explanation: By default, native VLAN frames are sent untagged on trunk ports.

49.

How many bytes are reserved in the Ethernet frame header for the 802.1Q VLAN tag?
A) 2 bytes
B) 4 bytes
C) 6 bytes
D) 8 bytes
Answer: B
Explanation: The 802.1Q VLAN tag adds 4 bytes to the Ethernet frame header.

50.

Which STP port state immediately forwards frames and does not participate in STP?
A) Blocking
B) Listening
C) Learning
D) Forwarding (PortFast)
Answer: D
Explanation: PortFast ports transition immediately to forwarding and do not participate in STP, designed for end devices.